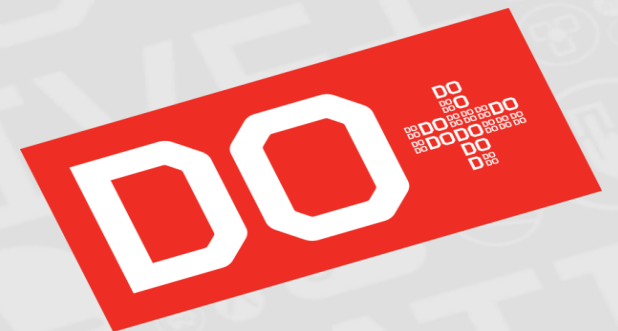




LXCA Configuration Patterns

David Roberts | Configuration Patterns– 4/15/2015

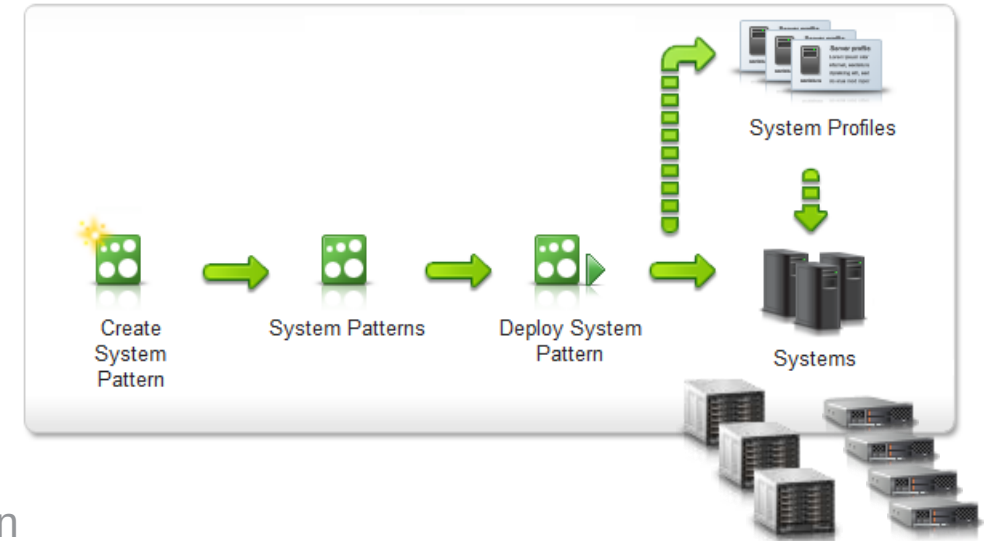


Configuration Patterns Introduction

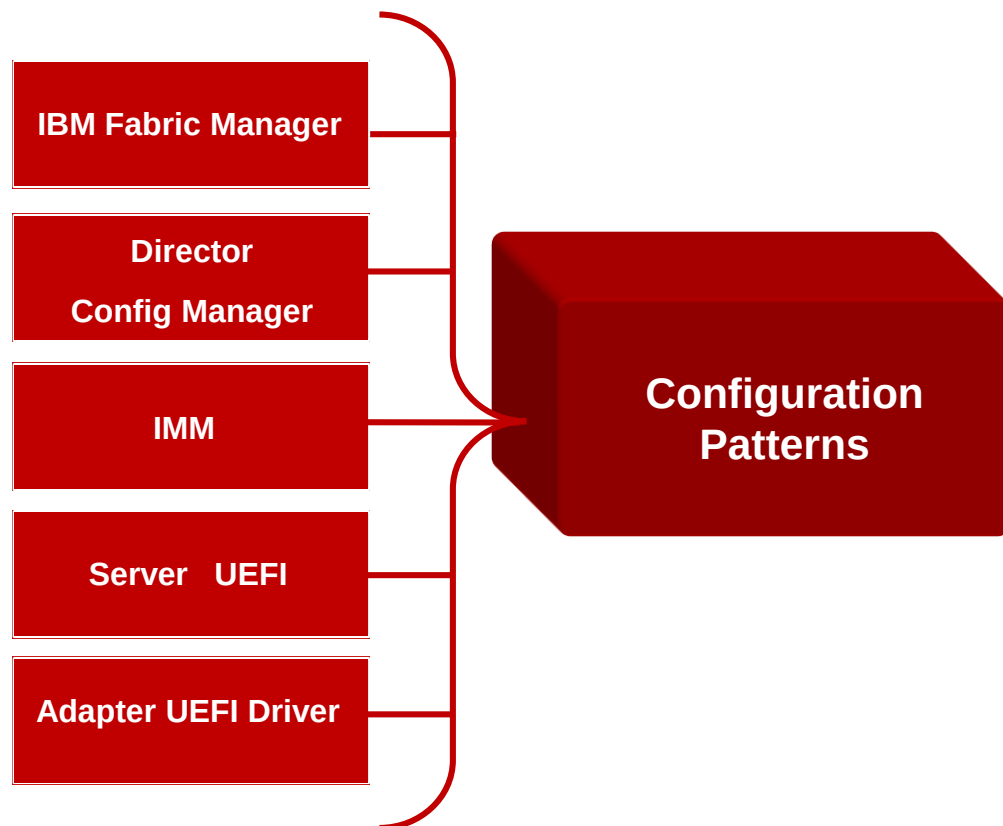
An integrated configuration management solution that simplifies the end-to-end task of provisioning System x hardware resources

Client Value

- Reduce time spent on hardware rollout
 - Deploy pre-OS configuration to multiple devices from a single pattern
 - Integrate device configuration into single interface and workflow
 - local storage, IMM, UEFI, I/O adapter and VNIC settings, virtual MAC/WWN addressing, SAN boot target
- Centralized pattern based management of configuration lifecycle
 - Consistent configuration baseline
 - Change configuration of multiple devices from a single interface
- Enable support for pre-configuration of ordered hardware, before physical acquisition
 - Pre-provision hardware by deploying placeholder configurations



Configuration Patterns: Integrated Solution



- Consolidate operations into 1 unified solution that provides 1-to-many automation
- Guided approach to creating pools, patterns, and profiles
- Create reference patterns by capturing from existing hardware
- Tight binding of configurations to managed devices
- Propagation of pattern configuration changes
- Assistance with device/pattern differences

Configuration Patterns: Differences from IBM FSM version

- Chassis Pattern support not available
 - Chassis component (Nodes, CMM, IOMs, etc) network configuration available Chassis Component Network Configuration UI
- Compute node failover support not available
 - Manual failover supported via profile deactivate / reactivate
- In order to clarify confusion related to server profile actions, some terminology changes have been implemented
 - Deploy profile is now called “Activate Profile”
 - Unassign profile is now called “De-activate Profile”
 - Profile status states are now ‘Active’ and ‘Inactive’
 - Deploy Pattern action remains unchanged

Configuration Patterns: New Features

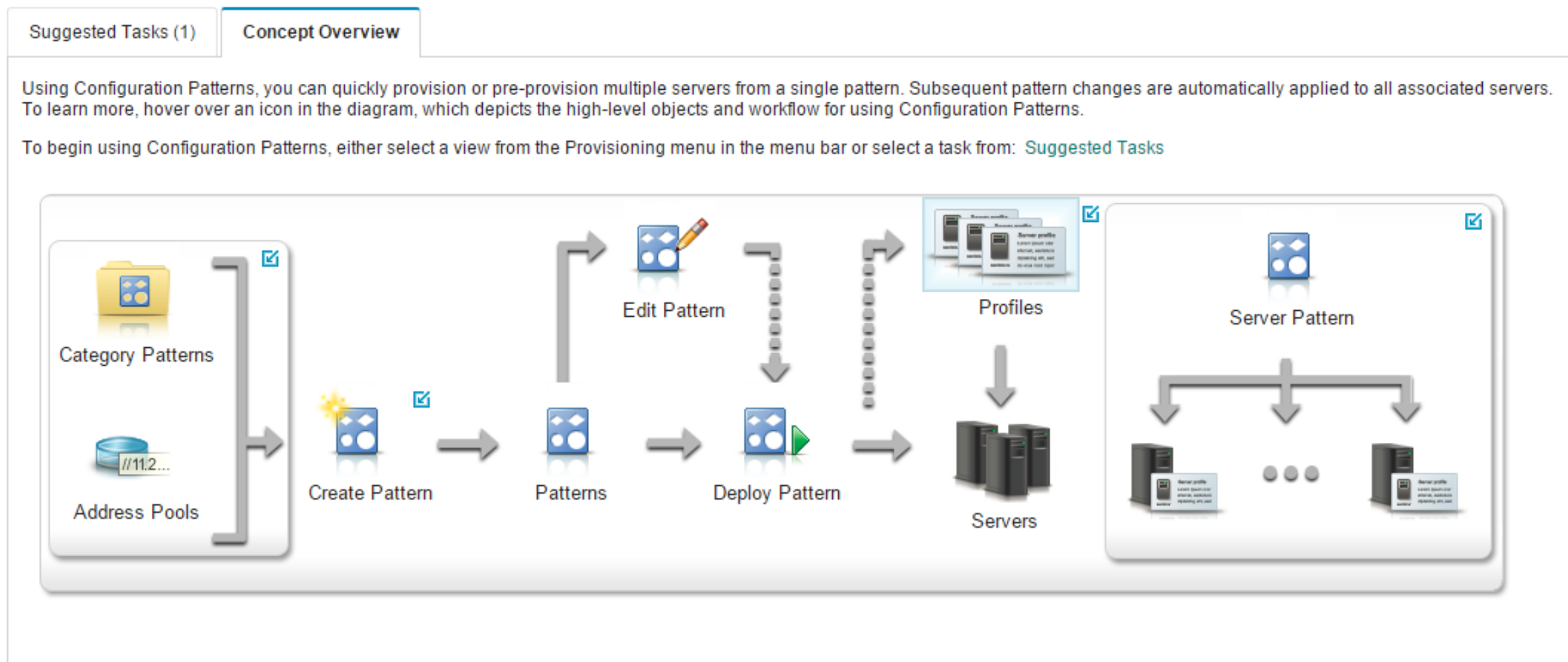
- User defined Ethernet & Fibre address pools
 - User supplied definition of address ranges
- Virtual address usage reporting
 - Expand support to include a view of the active ports and assigned addresses
- Pattern compatibility confirmation before deployment
 - Provide user with ability to review and confirm compatibility with servers before proceeding with deployment
- Parallel Deployment
 - Process queue in parallel, improving deploy processing time
- Pre-defined UEFI Patterns
 - Provide client with predefined UEFI patterns that match ATS recommendations for common workloads (Low Latency, Virtualization, etc)
- Import & Export of Patterns
 - REST API support for importing and exporting server pattern and associated category patterns

Configuration Patterns: Hardware Enhancements

- Broaden compute node configuration extending support to rack based systems
 - Integrate rack servers with Flex node workflow, avoid dissimilar experiences
 - Support RAID, I/O Adapter, & IMM/UEFI Settings
 - Address virtualization not supported
- Supported Servers:
 - x3550 M5, x3650 M5, x3960 X6
- Provide support for key chassis switch configuration
 - Add key switch configuration functions for Lenovo Flex chassis Ethernet switches
 - Focused on functions that require coordination of ITE level and Chassis switch configuration
 - eg. vNIC/UFP & VLAN ID tags

Configuration Patterns: Getting Started

- Introduction to concepts, including user objects and workflow



Configuration Patterns: New Server Pattern Wizard

- User starts by selecting either create from scratch or from existing server

New Server Pattern Wizard

General Local Storage I/O Adapters Boot Firmware Settings

Before you begin this wizard

▼ Select a starting point

 Create a new pattern from an existing server

 Create a new pattern from scratch

Back Next Save Save and Deploy Cancel

Configuration Patterns: New Server Pattern Wizard Step 1

- Select server type (Flex vs Rack) and enter pattern name

New Server Pattern Wizard

General Local Storage I/O Adapters Boot Firmware Settings

? Before you begin this wizard

► Starting point: Create a new pattern from scratch.

Specify pattern form factor

Form Factor: ? Flex Compute Node ▼

Specify pattern name and description

* Name: LXCA DST

Description (limit of 500 characters)

Back Next Save Save and Deploy Cancel

Configuration Patterns: New Server Pattern Wizard Step 2

- Local Storage: Basic RAID Boot volume, keep existing, or disable for SAN boot

New Server Pattern Wizard

General Local Storage I/O Adapters Boot Firmware Settings

Define the storage configuration that is to be applied to target servers when this pattern is deployed.

Select local storage configuration

 Specify storage configuration

 Keep existing storage configuration on target

 Disable local disk

This option makes no changes to existing storage configuration on target servers when this pattern is deployed.

Back Next Save Save and Deploy Cancel

Configuration Patterns: New Server Pattern Wizard Step 3

- Define IO Adapter configuration

New Server Pattern Wizard

General

Local Storage

I/O Adapters

Boot

Firmware Settings

?

 If desired, you can modify adapter addressing and define additional adapters to match the hardware that you expect to configure with this pattern.

I/O adapter addressing: ?

Burned In

Virtual



Non-Scalable Compute Node ▾

☐ Advanced Settings

All Actions ▾

☐ Location	Type	PCI Slot	Configuration Pattern	I/O Addressing	Description
☐  Compute Node					
☐ Add I/O Adapter					No adapter defined

Back

Next

Save

Save and Deploy

Cancel

Configuration Patterns: IO Adapter

- Select IO Adapters and define settings via Port Pattern

Add I/O Adapter 1 or LOM

PCI Slot: ▼

▶ Selected Adapter: Embedded 10Gb Virtual Fabric Ethernet Controller (LOM)

Select initial patterns. (?)

Initial port pattern: (?) ▼  

Configuration Patterns: Port Pattern

■ Configure Ethernet & Fibre Port Settings

New Port Pattern

Specify name and description

* Name:

Description:

Specify the adapter and port compatibility. ?

Target adapter type:

Target port operational mode:

Target port protocols:

Port extended settings pattern: ? 

☐ Apply corresponding settings to the chassis switch internal ports where applicable

■ Port Virtualization Settings

— vNIC, UFP Mode

■ Storage Protocols

— FCoE, iSCSI enablement

■ Extended Settings

— Learned adapter settings for advanced Ethernet and Fibre Channel settings

Configuration Patterns: Port Pattern, Chassis Ethernet Switch Support

- New in LXCA, Support for configuring chassis Ethernet switch internal port settings

New Port Pattern

☒ Apply corresponding settings to the chassis switch internal ports where applicable

i The following switch related settings can only apply to 4 types of switches. [Show Details](#)

UFP Function Settings Advanced Settings

Function	Enable	Mode	*Bandwidth (%)	VLAN Settings
1 st Function	☒	Trunk	25 - 100	Native VLAN 101 VLANs
2 nd Function	☒	Trunk	25 - 100	Native VLAN 102 VLANs
3 rd Function	☒	Trunk	25 - 100	Native VLAN 103 VLANs
4 th Function	☒	Trunk	25 - 100	Native VLAN 104 VLANs

Create Cancel

- Supported on Lenovo branded chassis switches:
 - EN4093R, CN4093, SI4093 & SI4091
 - Integrates configuration that requires coordination node and chassis switch configuration
 - Including, UFP Port Provisioning, VLAN Membership, flow control and failover

Configuration Patterns: IO Adapter View

- Adapter, port patterns and enabled ports displayed

New Server Pattern Wizard

General Local Storage **I/O Adapters** Boot Firmware Settings

? If desired, you can modify adapter addressing and define additional adapters to match the hardware that you expect to configure with this pattern.

I/O adapter addressing: ? **Burned In** Virtual

+ | Non-Scalable Compute Node | Advanced Settings | | All Actions

☐ Location	Type	PCI Slot	Configuration Pattern	I/O Addressing	Description
☐ Compute Node					
☐ LOM Fabric Connector	Virtual Fabric	1			Embedded 10Gb Virtual Fabric Ethernet Controller (LOM)
☐ Port 1	Virtual Fabric		UFP Mode	Burned-in Addresses	Physical Virtual Fabric port
☐ Function 1 — NIC				Burned-in Addresses	Virtual Ethernet port
☐ Function 2 — NIC				Burned-in Addresses	Virtual Ethernet port
☐ Function 3 — NIC				Burned-in Addresses	Virtual Ethernet port
☐ Function 4 — NIC				Burned-in Addresses	Virtual Ethernet port
☐ Port 2	Virtual Fabric		UFP Mode	Burned-in Addresses	Physical Virtual Fabric port
☐ Add I/O Adapter					No adapter defined

Back Next Save Save and Deploy Cancel

Configuration Patterns: Address Virtualization

■ Enable Virtualization of IO Hardware Addresses

Edit Virtual Addressing

Ethernet (MAC) virtual addressing **Enabled**

Ethernet (MAC) address pool:
Lenovo MAC Addresses ▼

Address range:
Default Lenovo Range 1 ▼ (576 out of 7281 addresses are allocated)

Fibre Channel (WWN) virtual addressing **Enabled**

Fibre Channel (WWN) address pool:
Lenovo WWN Addresses ▼

Address range:
Default Lenovo Range 1 ▼ (288 out of 466032 addresses are allocated)

Save Cancel

■ Virtual addressing

- Provides ability to re-assign or “virtualize” IO Adapter Ethernet and Fibre port addresses, overriding burned in addresses

■ Enables SAN Boot manual failover

- By assigning addresses to a profile instead of burned in, profiles can be deactivated and activated to a new node upon node failure

■ New in LXCA, support for defining custom address pools

- Users can now define custom addresses pool sizes & starting addresses

Configuration Patterns: New Server Pattern Wizard Step 4

■ Boot order

The screenshot shows the 'New Server Pattern Wizard' at Step 4, 'Boot'. The wizard has five tabs: 'General', 'Local Storage', 'I/O Adapters', 'Boot' (selected), and 'Firmware Settings'. Below the tabs, a text box states: 'This pattern can be used to configure the boot order for Legacy Only boot environments and SAN boot targets for UEFI or Legacy environments.' Below this, the 'System boot mode' section has five radio button options: '?', 'UEFI Only Boot', 'UEFI First, Then Legacy', 'Legacy Only Boot', and 'Keep existing boot mode' (which is selected). Below the radio buttons are three sub-tabs: 'Primary Boot Order' (selected), 'Wake on LAN (WoL) Boot Order', and 'SAN Boot'. A blue information box with a white 'i' icon contains the text: 'Boot order can only be configured if the Legacy Only Boot option is selected as the system boot mode.' and a 'Show Details' link. At the bottom of the wizard are five buttons: 'Back', 'Next' (highlighted with a dashed border), 'Save', 'Save and Deploy', and 'Cancel'.

- Most useful for legacy boot
 - UEFI Oses typically modify the boot order as part of install
 - Defining boot order most useful for legacy boot installations

Configuration Patterns: New Server Pattern Wizard Step 5

■ Firmware Settings

New Server Pattern Wizard

General Local Storage I/O Adapters Boot **Firmware Settings**

Integrated Management Module (IMM) and Server Firmware Settings (UEFI)

Select existing or create new category patterns as desired to include in this server pattern.

Category	Pattern		
System Information: ?	— No Pattern Selected —		
Management Interface: ?	— No Pattern Defined —		
Device And IO Ports: ?	— No Pattern Defined —		
Extended IMM: ?	— No Pattern Defined —		
Extended UEFI: ?	— No Pattern Selected —		

[Learn more about Extended Patterns](#)

M5 High Performance
M5 Low Latency
M5 Virtualization
x240 High Performance
x240 Low Latency
x240 Standard Workload
x240 Transactional Workload
x240 Virtualization
x6 High Performance

Back Next Save Save and Deploy Cancel

■ Patterns for modifying:

- System info, including location & device name
- Management Interface, IMM IP, Hostname
- Device & IO Ports, Console redirection & SOL
- Extended IMM & UEFI Settings

Configuration Patterns: Extended UEFI Patterns

- New in LXCA, Pre-defined UEFI Patterns

Edit Extended UEFI Pattern - M5 High Performance

Turbo Mode	☒ Enabled
Processor Performance States	☒ Enabled
C-States	☐ Disabled
C1 Enhanced Mode	☐ Disabled
Hyper-Threading	☒ Enabled
Execute Disable Bit	☒ Enabled
Trusted Execution Technology	☐ Disabled
Intel Virtualization Technology	☒ Enabled
Hardware Prefetcher	☒ Enabled
Adjacent Cache Prefetch	☒ Enabled
DCU Streamer Prefetcher	☒ Enabled
DCU IP Prefetcher	☒ Enabled
DCA	☒ Enabled
Energy Efficient Turbo	☐ Disabled

- Pre-defined, ATS provided best practices, optimized UEFI settings for typical workloads
 - Virtualization (hypervisors)
 - High Performance
 - Low Latency
 - Transactional workloads

Configuration Patterns: Deploy Server Compatibility Verification

Deploy Server Pattern - LXCA DST

Deploy the server pattern to one or more individual servers or groups of servers (for example, a chassis). During deployment, one server profile is created for each individual server.

* Pattern To Deploy: LXCA DST

* Activation ?

- ☒ Full — Activate all settings and restart the server now.
☐ Partial — Activate IMM settings but do not restart the server. UEFI and server settings will be active after the next restart.
☐ Deferred — Generate a profile with the settings for review, but do not activate settings on the server.

Choose one or more servers to which to deploy the selected pattern.

☐ Show Empty Bays

Any Deploy Status

Filter

☐	Name	Rack Name/Unit	Chassis/Bay	Deploy Status
☒	Lenovo x440	Unassigned / Unassigned	SN#Y030BG21E01C, Bay 1,2	✓ Ready
☐	IBM x880 (Pri)	Unassigned / Unassigned	SN#Y030BG21E01C, Bay 3,4,5,6	⚠ Not Supported More Details
☐	Lenovo x240 M5	Unassigned / Unassigned	SN#Y030BG21E01C, Bay 7	⚠ Not Supported More Details
☐	Lenovo x240	Unassigned / Unassigned	SN#Y030BG21E01C, Bay 8	⚠ Not Supported More Details

Deploy

Cancel

- Ensures server has minimum required firmware levels
- Server is available and not locked by another provisioning activity
- Validates selected Adapter Port Pattern is compatible with server

Configuration Patterns: Job Status

■ Monitor deploy process via Jobs UI



Deployment request was submitted.

Job "Server Profile activation: Apr 13, 2015" has been created and started successfully. Changes are being propagated to the following servers or bays: Lenovo x440

You can monitor job progress from the Jobs pod in the banner above.

You can view the profile creation progress from the Server Profiles link that is located under the Provisioning menu in the menu bar. Profiles will not show up in the Server Profiles table until the profile has been created.

[Jump to Jobs page](#)[Jump to Profiles page](#)[Return to Patterns page](#)

Jobs

Jobs are longer running tasks performed against one or more target systems. After selecting a job, you can choose to cancel it, delete it, or obtain details.



Show:

☐ Job	Status	Start	Complete	Targets
☐ Profile activation for server Lenovo	50%	April 13, 2015 at 08:04:08		
☐ Server Profile activation: Apr 13, 2015	50%	April 13, 2015 at 08:03:50		
☐ Deploy Server Settings	50%			SN#Y030BG21E01C
☐ Manage job for 97c28df7541b465	Complete	April 13, 2015 at 07:53:19	April 13, 2015 at 07:59:01	SN#Y030BG21E01C Management USERID
☐ Unmanage job for F44E92339683	Complete	April 13, 2015 at 07:49:51	April 13, 2015 at 07:49:52	Not Available Management USERID

Summary:

Name:	Server Profile activation: Apr 13, 2015
Time Started:	April 13, 2015 at 08:29:07
Overall Status:	Complete
Target:	
Created By:	USERID

Target Results: With Errors: 0 Running: 0 Completed: 1

Target	Message
SN#Y030BG21E01C: Lenovo x440: Bay 1-2	Deploy Server Settings

Close

Configuration Patterns: Troubleshooting

▼ Summary:

Name:	Server Profile activation: Apr 13, 2015
Time Started:	April 13, 2015 at 08:03:50
Overall Status:	✖ Stopped With Error
Target:	
Created By:	USERID

▼ Target Results: With Errors: 1 Running: 0 Completed: 0

Target	Message
SN#Y030BG21E01C: Lenovo x440: Bay 1-2	Deploy Server Settings

▼ Log

Close

- Most failures are captured and logged via the job log
 - Errors communicating with server
 - Failure to restart server
 - Incompatible settings
- Detailed failure data captured via FFDC captures
 - Capture ID: 5101
 - Includes logs, & diagnostic data
 - Archive of persisted user data, including created patterns, profiles & pools

 Download All Service Data

 Download logs only



All actions ▼

Filter

	Date and Time ▲	File	Description	
☐	Apr 13, 2015, 8:04:08 AM	INFO_5 (ffdcCapture_11001-5.zip)	com.lenovo.lxca.job.ActionJob DumpID 11001	
☐	Apr 13, 2015, 8:07:48 AM	INFO_6 (ffdcCapture_11001-6.zip)	com.lenovo.lxca.job.ActionJob DumpID 11001	
☒	Apr 13, 2015, 8:10:21 AM	FFDC_3 (ffdcCapture_5101-3.zip)	com.lenovo.lxca.server.servlets.ffdc.FFDCDumpArchiveServlet DumpID 5101	▲